Full Stack Engineer — Java, Quarkus / Spring Boot & Angular | AI-Driven Microservice Architectures

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PROFESSIONAL SUMMARY

Full Stack Engineer with 3 years of experience at Blue Reply on mission-critical digital platforms for consumer credit (**Java**, **Quarkus**, **Spring Boot**, **Angular**). The level of responsibility has grown over time: from end-to-end delivery of features on existing platforms — calculation engines, anti-fraud checks, CMS modules — to designing greenfield **microservices** with modern **patterns** (hexagonal architecture, **MicroProfile** resilience, **GenAI**/**OCR** solutions), up to owning the entire **microservice** architecture (**Quarkus**) of a multichannel platform (37 services): surveying hundreds of **APIs**, C4 modeling, designing external integrations, and defining team standards. Cross-functional experience with **API** Management (WSO2, **3scale**), performance optimization, and **event-driven** architectures (**Kafka**), from design through to production release via **CI/CD**.

WORK EXPERIENCE

IT Consultant | Full Stack Engineer — Blue Reply S.r.l., Milan · 09/2023 – present

Full stack development of web platforms and **microservices** for major Italian consumer-credit and mobility groups (including Agos, Findomestic, Fiditalia, and Drivalia), in **Agile**/**Scrum** teams, with **CI/CD** on GitLab and **Jenkins**, and daily use of **Claude Code** and **Cowork** as **AI-native** development and **automation** tools (**agentic coding**, **prompt engineering** applied to software).

Architectural design and microservices

- Designed and governed a pipeline of 7 specialized **Claude agents** (requirements/business, architecture, development, testing, **API**/**microservice** cost estimation), model routing by role (**Opus** for architecture/business, **Sonnet** for execution), **knowledge base** split into an index and sections, and **automatic** per-agent **token/cost logging** to decide **cost-optimization** changes.
- Implemented the suite of 3 **Quarkus microservices** for multichannel OTP authentication (sending and validation via SMS, email, and push) designed within the same model, with 8+ versions released in 9 months: type-safe **REST** clients generated from **OpenAPI** specifications, **MicroProfile** fault tolerance (timeout, retry, circuit breaker), custom bean validation, health checks, and a dedicated TLS truststore, with **APIs** exposed via WSO2 **API** Manager and **Red Hat 3scale**, deployed on **Red Hat OpenShift**.
- Led the architectural migration and modernization of a self-service platform for the customers of a leading consumer-credit institution (web and mobile app channels) toward a new **microservice** architecture: surveyed 150+ **REST** endpoints by cross-referencing the app and web channel **APIs** (raw extraction, identifying ~30 channel-exclusive endpoints), then remapped them onto 37 domain-designed **Quarkus microservices** across 3 application areas (customer records, cards, loans and financing) — ~100 endpoints exposed to the BFF across 12 management **microservices**, 120+ external-vendor endpoints across 19 dedicated adapters, and 20 internal endpoints across 6 adapters toward shared services. Development of the **microservices** adopts an internal **Claude Code** command framework that **scaffolds** adapters, BFX, and management services following team conventions, reducing manual boilerplate writing.
- Formalized the architectural model in 6 C4 diagrams (draw.io) and 40+ sequence diagrams (login, customer records, loans, cards, payment security, OTP, and back-office), published on Confluence with ~50 **microservices** documented in linked pages: these have become the shared reference point for the development team, client architects, and vendors.
- Designed integrations with 10 external systems and vendors (core banking, CRM, tracking, postal correspondence, document archiving, collections, digital signature and OTP providers, ACS 3D Secure, **message queue**) applying an adapter chain **pattern** with centralized authentication, isolating every external protocol (**SOAP**, **REST**, JSON-RPC) behind a uniform internal contract.
- Defined and documented the architectural conventions adopted by the team — parallel BFFs for web and mobile, management services per domain, internal adapters toward shared services, event processing on **Kafka**, naming and namespacing — and industrialized documentation production by creating a **Cowork** skill that **automatically** compiles the C4 draw.io diagrams from a JSON architecture model, cutting diagram updates down to a few minutes.

GenAI and document OCR

- Built a **GenAI** solution for data extraction from identity documents: a **Python** prototype **fine-tuning** GPT-4o vision (**OpenAI API**) on synthetic datasets generated from facsimiles, with an agent that recognizes the document type and invokes the specialized model via **tool calling** (strict JSON schema); later ported to an **Angular** 18 + **OpenAI** SDK web app with parallel tool calls, Markdown output, and token-usage monitoring.

Payment integrations — REST exposure layer

- Developed the Click to Pay payment flow end-to-end on the **REST** exposure layer: 12 **Camel** routes and 22 **Java** processors for login/session, payment initiation, case-file retrieval, barcode tracking, and receipt download, in continuous release through today.

Performance optimization — content distribution service

- Delivered 14 service releases (v1.1.21 → v1.1.34) in 3 months using GitFlow, reducing load on the **API** gateway (WSO2) and the database by designing and implementing a versioned JSON content-caching system: a virtual status endpoint with a filename → version-date manifest, server-side Spring caching (synchronized @Cacheable, scheduled eviction with reconfigurable TTL), database protection against non-catalog id requests, and a download fallback for the web channel.

Loan origination platform and digital journey — full stack

- Developed the IBAN blacklist anti-fraud feature end-to-end — **REST** services and MyBatis DAOs on the **Java**/Spring backend (**JBoss EAP**), **Angular** search and configuration pages on the frontend — together with email/mobile anti-fraud checks with audit reporting and customer IBAN traceability.
- Built the threshold and income-calculation configuration pages with the related evaluation logic, the scheduled processing jobs, and the remote sales process with signed contract, over 12 months across 25+ releases (v5.1.x).
- Developed the routing engine for existing-customer digital-journey applications (outcome handling, pre-loading diversion, repayment-plan remodulation) and the installment-calculation services, integrating customer records via **OpenAPI** client and **legacy SOAP**, with SPID and privacy consent — 340+ commits in 6 months across an **Angular** 14 frontend and a **Spring Boot microservice**.

Internal CMS — full stack

- Built the platform's CMS module from scratch: an **Angular** content-creation component (dynamic forms, partner/channel targeting, attachments) and filtered search, with **Spring Boot**/MyBatis **APIs** for search, detail, and upsert; resolved 10+ defects in UAT.

Software Tester (Master's thesis internship) — TeamSystem S.p.A., Catania · 05/2023 – 08/2023

- Defined best practices for isolated, reproducible testing in the **DevOps** pipeline: database reset between runs and third-party service simulation (test doubles).
- Designed and built a **containerized** load-testing pipeline (**Docker**, Makefile) orchestrating JMeter and **Prometheus**/cAdvisor monitoring, with a **Python** script for **automatic** reporting (latency, CPU, and memory charts in PDF); a single make command reproduces the entire test across different load scenarios and resource limits.

TECHNICAL SKILLS

Languages: Java (8–17), **TypeScript**, **JavaScript**, SQL, **Python**, Dart

Backend: Quarkus, **MicroProfile** (Fault Tolerance, **REST** Client), **Spring Boot** 2/3, **Apache Camel**, **microservices**, **REST** and **SOAP**, **OpenAPI**/Swagger, OAuth2, Spring Cache, **JPA**/**Hibernate**, MyBatis, **Node.js** (Express), Apache **Kafka**

Frontend: Angular 14–18, NgRx, RxJS, HTML5, CSS3/SCSS, Bootstrap, Flutter, React (basic knowledge applied to personal projects)

Architecture: C4 modeling (draw.io) and sequence diagrams, solution design and **API** cataloging, domain decomposition, BFF/BFB **patterns**, management, adapter, event processing on **Kafka**, **API** Management (WSO2 **API** Manager, **Red Hat 3scale**), JWT

DevOps & tools: Git, GitLab **CI/CD**, **Jenkins**, **Docker**, **Red Hat OpenShift**, **Maven**, Gradle, **Prometheus**, Jira, Confluence, **Claude Code**, **Cowork**

Testing: JUnit, JMeter, Postman, SoapUI

AI-Driven Development: Claude Code and **Cowork** in daily production use (**agentic coding**, **prompt engineering** applied to software, creation of custom skills and **automations**), **LLM** integration (**OpenAI** SDK), document **OCR**

Methodologies: Agile/**Scrum**, code review, **CI/CD**, technical documentation

CERTIFICATIONS

Flutter Certified Application Developer — Android ATC (2024)

EDUCATION

Master's Degree in Computer Engineering (LM-32) — University of Catania · 2021 – 2023 · 110/110 with honors

Thesis: “Test Analysis in the **DevOps** Pipeline” (internship at TeamSystem S.p.A.)

Bachelor's Degree in Computer Engineering (L-8) — University of Catania · 2018 – 2021 · 104/110

LANGUAGES

Italian (native) · English (B2)

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